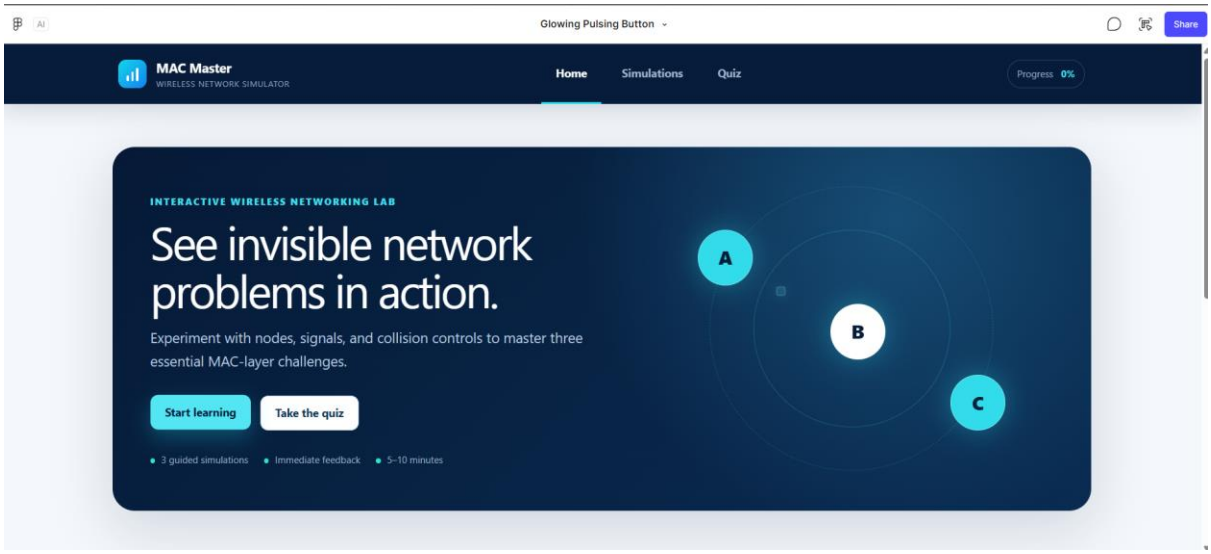
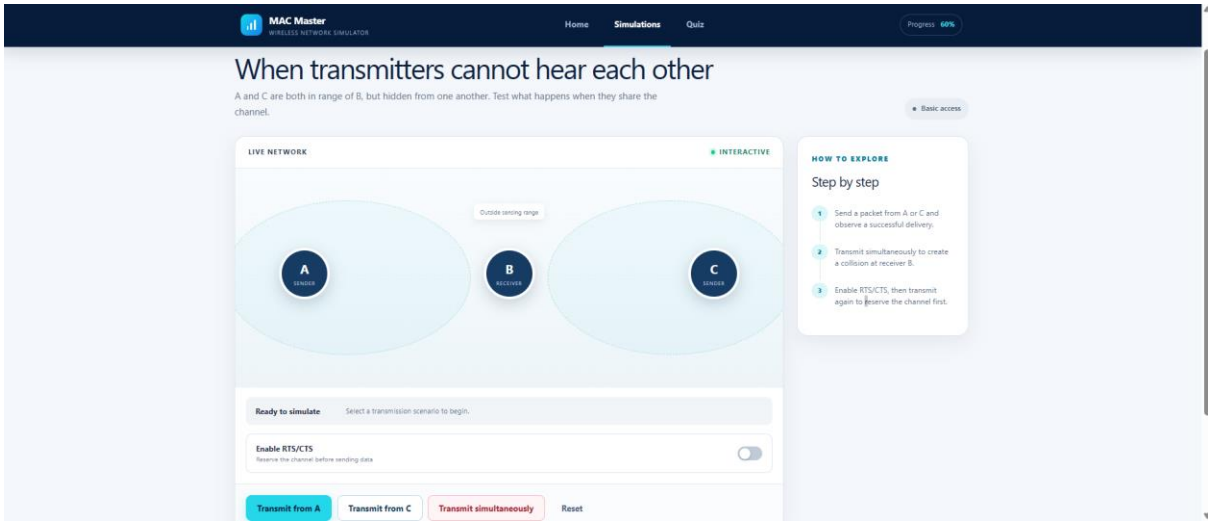
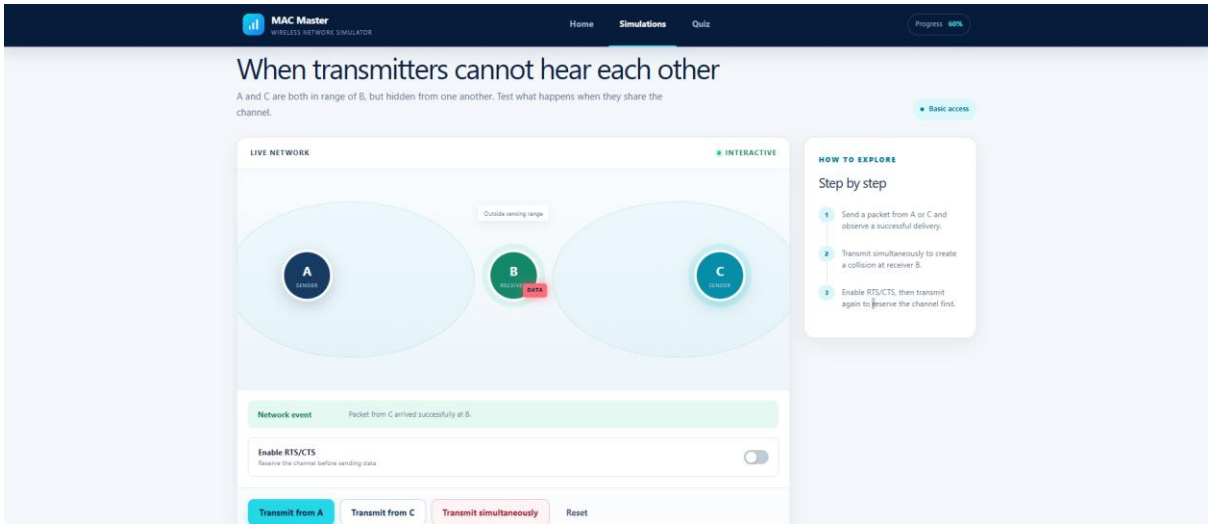
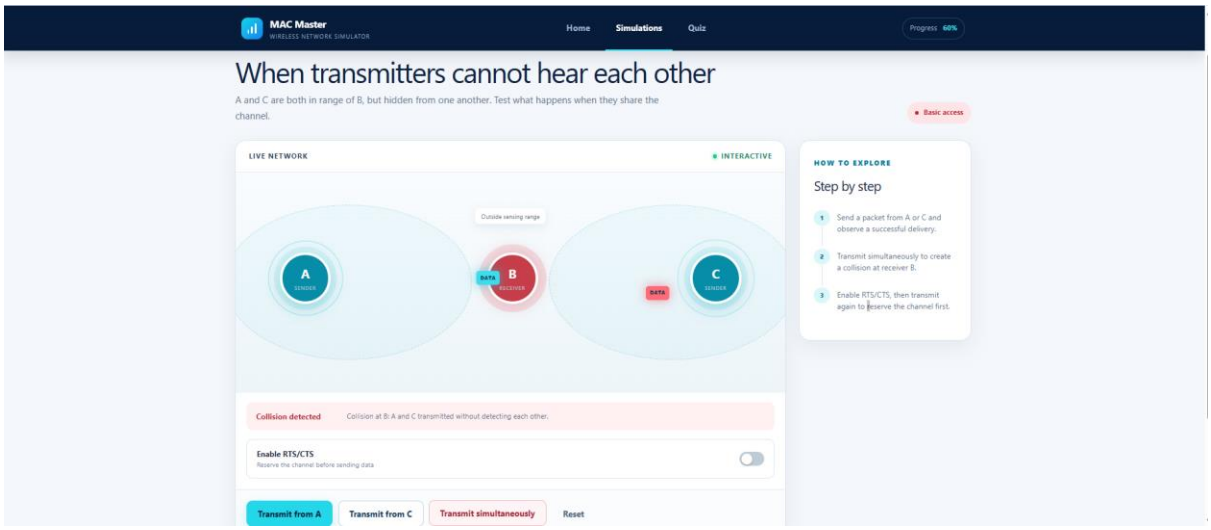
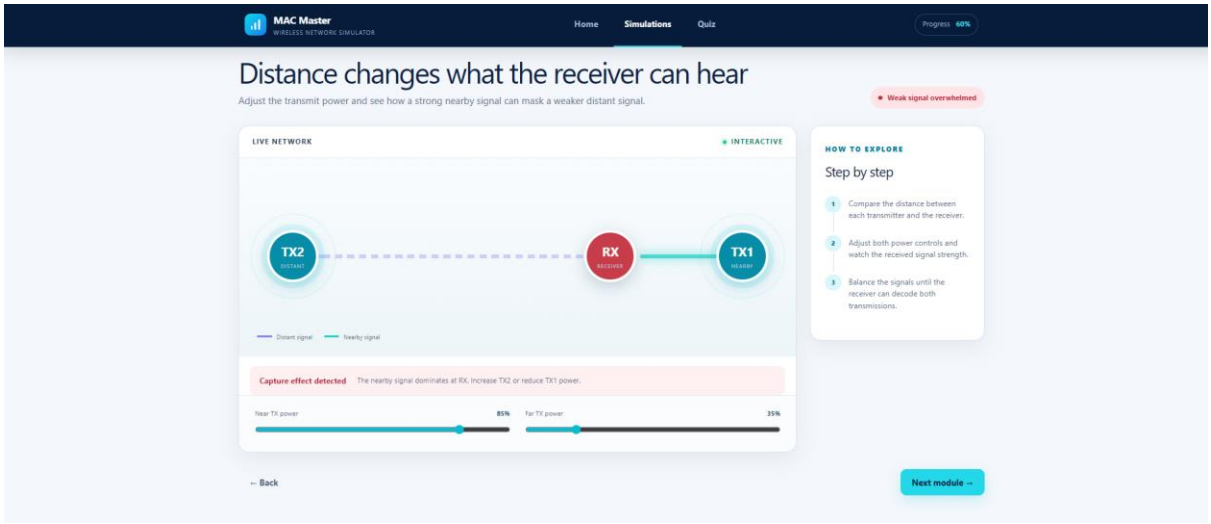
ACTIVITY **J. SAI DHEERAJ**

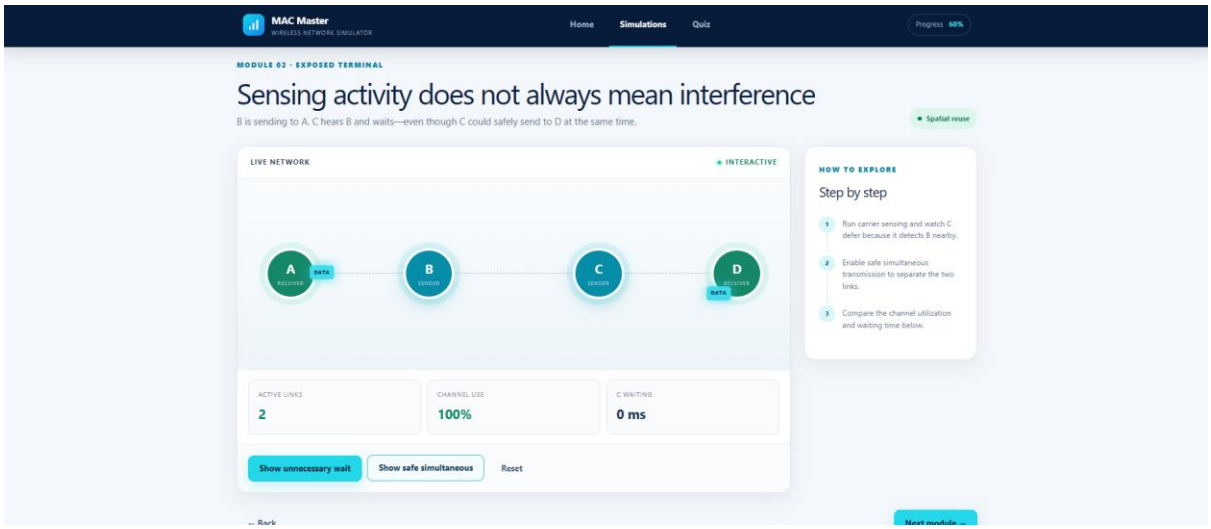


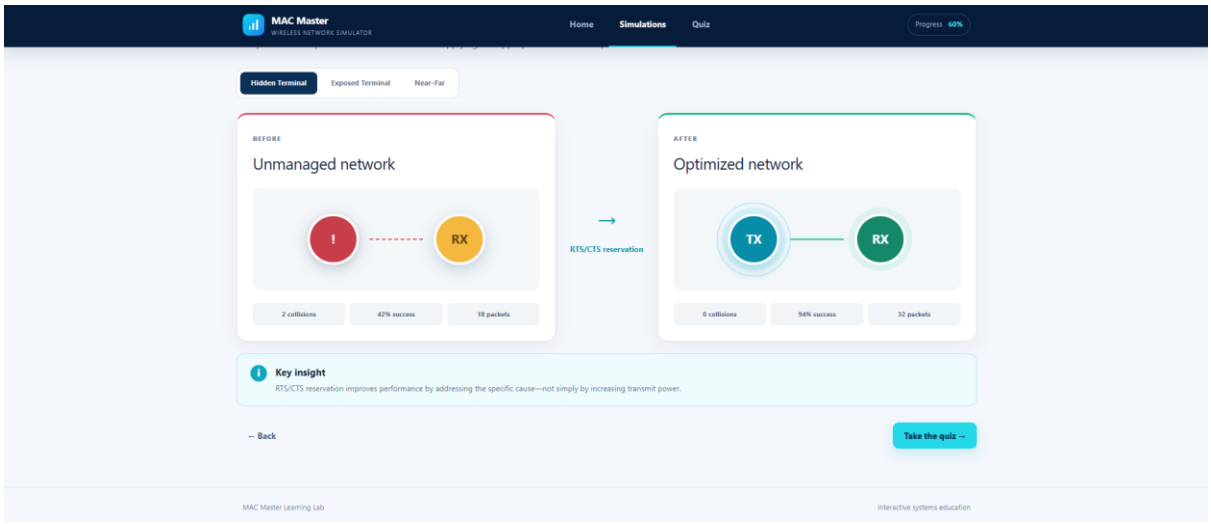


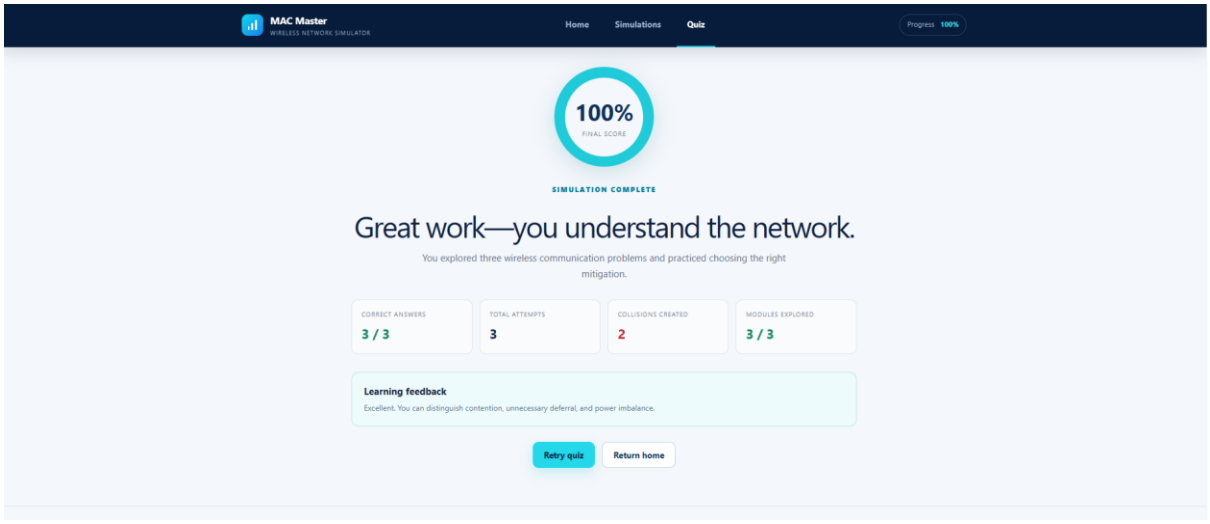



MAC Master
WIRELESS NETWORK SIMULATOR
Home
Simulations
Quiz
Progress 60%
When transmitters cannot hear each other
A and C are both in range of B, but hidden from one another. Test what happens when they share the channel.
Basic access
LIVE NETWORK
INTERACTIVE
Outside sensing range
A
SENDER
DATA
B
RECEIVER
DATA
C
SENDER
HOW TO EXPLORE
Step by step
1 Send a packet from A or C and observe a successful delivery.
2 Transmit simultaneously to create a collision at receiver B.
3 Enable RTS/CTS, then transmit again to reserve the channel first.
Collision detected
Collision at B: A and C transmitted without detecting each other.
Enable RTS/CTS
Reserve the channel before sending data
Transmit from A
Transmit from C
Transmit simultaneously
Reset


MAC Master
WIRELESS NETWORK SIMULATOR
Home
Simulations
Quiz
Progress 60%
Distance changes what the receiver can hear
Adjust the transmit power and see how a strong nearby signal can mask a weaker distant signal.
Weak signal overwhelmed
LIVE NETWORK
INTERACTIVE
TX2
DISTANT
RX
RECEIVER
TX1
NEARBY
Distant signal
Nearby signal
HOW TO EXPLORE
Step by step
1 Compare the distance between each transmitter and the receiver.
2 Adjust both power controls and watch the received signal strength.
3 Balance the signals until the receiver can decode both transmissions.
Capture effect detected
The nearby signal dominates at RX. Increase TX2 or reduce TX1 power.
Near TX power
85%
Far TX power
35%
← Back
Next module →


MAC Master
WIRELESS NETWORK SIMULATOR
Home
Simulations
Quiz
Progress 60%
MODULE 02 · EXPOSED TERMINAL
Sensing activity does not always mean interference
B is sending to A. C hears B and waits—even though C could safely send to D at the same time.
Spatial reuse
LIVE NETWORK
INTERACTIVE
A
RECEIVER
DATA
B
SENDER
C
SENDER
D
RECEIVER
DATA
HOW TO EXPLORE
Step by step
1 Run carrier sensing and watch C defer because it detects B nearby.
2 Enable safe simultaneous transmission to separate the two links.
3 Compare the channel utilization and waiting time below.
ACTIVE LINKS
2
CHANNEL USE
100%
C WAITING
0 ms
Show unnecessary wait
Show safe simultaneous
Reset
← Back
Next module →


MAC Master
WIRELESS NETWORK SIMULATOR
Home
Simulations
Quiz
Progress 60%
Hidden Terminal
Exposed Terminal
Near-Far
BEFORE
Unmanaged network
!
RX
2 collisions
42% success
18 packets
RTS/CTS reservation
AFTER
Optimized network
TX
RX
0 collisions
94% success
32 packets
Key insight
RTS/CTS reservation improves performance by addressing the specific cause—not simply by increasing transmit power.
← Back
Take the quiz →
MAC Master Learning Lab
Interactive systems education


MAC Master
WIRELESS NETWORK SIMULATOR
Home
Simulations
Quiz
Progress 100%
100%
FINAL SCORE
SIMULATION COMPLETE
Great work—you understand the network.
You explored three wireless communication problems and practiced choosing the right mitigation.
CORRECT ANSWERS
3 / 3
TOTAL ATTEMPTS
3
COLLISIONS CREATED
2
MODULES EXPLORED
3 / 3
Learning feedback
Excellent. You can distinguish contention, unnecessary deferral, and power imbalance.
Retry quiz
Return home